MANIDIPA SARKAR, Ph.D.

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Research Interests

- Ph.D. scholar in Electronics and Communication Engineering with research interests in wireless communications, massive MIMO, statistical signal processing, delayed channel state information, and channel aging. Focusing on developing efficient optimization algorithms for next-generation wireless communication networks.

Education

2023 – Present — **Ph.D in Wireless Communication, NIT Raipur**
Ph.D supervisor: Dr. Siddharth Deshmukh
Thesis title: *Algorithms for performance improvement of FDD operated massive MIMO network with delayed CSIT.*

2019 – 2021 — **M.Tech, Electronics and Communication Engineering, KIIT University** .
M.tech supervisor: Dr. Sarita Nanda
Thesis title: *Application of Adaptive Signal Processing Techniques for Channel Estimation of MIMO-NOMA Systems.*
CGPA: 9.07.

2014 – 2018 — **B.Tech. Electronics and Instrumentation Engineering, KIIT University.**
CGPA: 8.19.

Research Publications

Journal Articles

1. M. Sarkar and S. Deshmukh, "Decentralized online rate optimization in fdd massive mimo system with delayed csi feedback," *IEEE Transactions on Vehicular Technology*, vol. 75, no. 9, pp. 20 701–20 711, 2026. DOI: 10.1109/TVT.2026.3693346
2. M. Sarkar, A. Nayak, S. Nanda, and S. Patnaik, "Non-orthogonal multiple access (noma) channel estimation for mobile & plc-vlc based broadband communication system," *International Journal of Sensors, Wireless Communications and Control*, vol. 15, no. 1, pp. 30–39, 2025.
3. S. Sahoo, M. Sarkar, H. K. Sahoo, and S. Nanda, "Multiuser hybrid precoder design using logarithmic hyperbolic filtering for millimeter wave communication systems," *Wireless networks*, vol. 30, no. 1, pp. 139–150, 2024.
4. M. Sarkar, S. Sahoo, and S. Nanda, "Channel estimation of non-orthogonal multiple access systems based on l 2-norm extreme learning machine," *Signal, Image and Video Processing*, vol. 16, no. 4, pp. 921–929, 2022.

Conference Proceedings

1. M. Sarkar and S. Deshmukh, "Downlink rate maximization in multicell fdd massive mimo system with delayed csi feedback," in *2026 National Conference on Communications (NCC)*, 2026, pp. 25–30. DOI: 10.1109/NCC68160.2026.11478908

2 M. Sarkar and S. Deshmukh, "Downlink rate maximization in fdd operated massive mimo system with delayed csi feedback," in *2024 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, 2024, pp. 1–6. DOI: 10.1109/ANTS63515.2024.10898797

3 M. Alayu, M. Sarkar, and S. Nanda, "5g channel estimation of noma-ofdm system using arctangent lms algorithm," in *2023 2nd International Conference on Computational Systems and Communication (ICCSC)*, 2023, pp. 1–4. DOI: 10.1109/ICCSC56913.2023.10143017

4 A. Nayak, M. Sarkar, S. Sahoo, and S. Nanda, "Channel estimation of mimo-noma system based on a variable forgetting factor recursive least square algorithm," in *2021 IEEE 2nd International Conference on Applied Electromagnetics, Signal Processing, Communication (AESPC)*, 2021, pp. 1–5. DOI: 10.1109/AESPC52704.2021.9708457

5 M. Sarkar, S. Sahoo, Y. K. Barapatre, and S. Nanda, "Proportionate-type algorithm for mimo-ofdm channel estimation and its fpga implementation," in *2021 IEEE 4th International Conference on Computing, Power and Communication Technologies (GUCON)*, 2021, pp. 1–5. DOI: 10.1109/GUCON50781.2021.9574025

6 M. Sarkar, S. Sahoo, and S. Nanda, "A sparse based adaptive channel estimator for wireless channel," in *2020 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*, 2020, pp. 472–475. DOI: 10.1109/WIECON-ECE52138.2020.9398009

Skills

Languages	Strong reading, writing and speaking competencies for English, Hindi, Bengali.
Coding	Python, MATLAB, LaTeX.
Web Dev	Html.